

 Did You Know?

Normalization helps organize data efficiently by reducing duplication and improving data consistency.



What is Normalization?

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It divides large tables into smaller related tables.



Why do we use Normalization?

- Reduce data redundancy
- Improve consistency
- Avoid update anomalies
- Save storage space
- Make databases easier to maintain

Normal Forms

Normal Form	Description
1NF	Remove repeating groups and store atomic values.
2NF	Remove partial dependency.
3NF	Remove transitive dependency.
BCNF	Every determinant must be a candidate key.

1 First Normal Form (1NF)

A table is in First Normal Form when every column contains only atomic (single) values and there are no repeating groups.

StudentID

Name

Subject

2 Second Normal Form (2NF)

A table is in Second Normal Form when it is already in 1NF and all non-key attributes depend on the whole primary key.

3 Third Normal Form (3NF)

A table is in Third Normal Form when it is already in 2NF and has no transitive dependency.

4 Boyce-Codd Normal Form (BCNF)

BCNF is a stronger version of 3NF. Every determinant in the table must be a candidate key.



Example

```
Student(ID, Name, Department)
```

```
Department(DepartmentID, DepartmentName)
```



Advantages of Normalization

- Less duplicate data
- Better consistency
- Easy updates
- Improved database performance
- Efficient storage

Common Mistakes

- Confusing 2NF and 3NF
- Ignoring functional dependency
- Creating unnecessary tables
- Using repeated values

Interview Questions

Q1: What is Normalization?

Normalization is the process of organizing database tables to reduce redundancy.

Q2: What is 1NF?

It removes repeating groups and stores atomic values.

Q3: What is 3NF?

It removes transitive dependency.

Q4: Why is Normalization important?

It improves consistency and reduces duplicate data.



Quick Revision

1NF Atomic Values

2NF No Partial Dependency

3NF No Transitive Dependency

BCNF Candidate Key Rule



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